



NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
BSc (Hons) Computing|ITB|EHN|Year-1
DIPLOMA IN COMPUTER SYSTEM DESIGN|-22.1
DIPLOMA IN SOFTWARE ENGINEERING|-22.1
DIPLOMA IN NETWORK ENGINEERING|-22.1
DATABASE MANAGEMENT SYSTEMS
2022-10-31

This paper contains **05** questions

Answer All questions.

Time: **THREE** hours.

This paper contains only 5 questions. Answer all questions. Other than this exam student knowledge is evaluated through practical coursework.

The student's practical knowledge was evaluated during the Practical Course Work. For the Practical course work, students must get above 40 marks to pass this module.

Question One (Underline the correct answer in this paper).
(Each question carries 02 Marks. $2 * 10 = 20$ Marks)

1. How do you map m:n relationship to a schema?
 - a) Create a new relation for a Single value attribute
 - b) Create a new relation by taking One Primary Key from only one entity
 - c) Map same as the 1:m relationship
 - d) Create a new relation by taking Primary Keys from both entities
2. Which command is used to modify columns in a table?
 - a) Create
 - b) Drop
 - c) Truncate
 - d) Alter
3. What is the correct statement to add a new column Email to Student Table?
 - a) ALTER TABLE Student WHERE ADD Email VARCHAR (10);
 - b) ALTER TABLE Student GROUP BY ADD Email VARCHAR (10);
 - c) CREATE TABLE Student WHERE ADD Email VARCHAR (10);
 - d) ALTER TABLE Student ADD Email VARCHAR (10);
4. What is the correct statement regarding removing all the data from the Employee table?
 - a) TRUNCATE TABLE Employee
 - b) DELETE * FROM Employee
 - c) TRUNCATE TABLE Student
 - d) DELETE FROM Student

5. What is Foreign Key?
- Same as Primary Key
 - Use to identify each row uniquely
 - Another name for Composite Primary Key
 - Primary Key of another table which helps to create a connection
6. CHECK constraints are used to _____
- Provide a default value for a column
 - Ensure that all values in a column are different
 - Uniquely identify each row/record in a database table
 - Ensure that all the values in a column satisfy certain conditions
7. Student (Sid, Sname, Saddress, Email, TP)
- Which of the following query would display all the students where the first character in the name start with 'i'?
- SELECT Sname FROM Student WHERE Sname like '_i_';
 - SELECT Sname FROM Student WHERE Sname like 'i%';
 - SELECT Sname FROM Student WHERE Sname like '_i%';
 - SELECT Sname FROM Student WHERE Sname like '%i_';
8. What is the main concept of EER?
- Super Class and Minimum Class Concept
 - Super Class and Maximum Class Concept
 - Minimum Class and Maximum Class Concept
 - Super Class and Sub Class Concept
9. What kind of data should be included in a Hospital Database?
- Data related to Students, Courses, Lecturers and Exams
 - Data related to Items, Suppliers, Branches and Food
 - Data related to Wards, Treatments, Doctors and Students
 - Data related to Wards, Patients, Treatments and Doctors
10. Employee (Eid, Ename, Salary, Deptno);
- What is the correct statement to update Salary by 10% of all the Employees?
- UPDATE Employee SET Salary = Salary + (Salary * 10);
 - UPDATE Client SET Salary = Salary + (Salary * 0.1);
 - UPDATE Employee SET Salary = Salary + (Salary * 0.1);
 - UPDATE Employee SET Salary = Salary * 0.1;

Question Two (20 Marks)

A training institute conducts several courses, it needs a database to be developed to store the information about courses, lecturers, participants etc. A course (ccode, ctitle, sdate, edate, cname,duration, cfee) is managed by several course directors (cdcode, cdname, cdtel,division). There are many subjects (scode, sname) in a course and one subject can teach in many courses. The lecturer id (LID) is used to trace the lecturer who did a particular subject in a course. There are many lecturers (Lid, title, Lname, Dept, Laddr), lecturing for a course and one lecturer can attend for many courses. For lecture payments, it is important to store the date and the duration of the lecturers attended in different courses. A course gets payments from many participants (pcode, pname, paddr, dob, gender) and a participant pay for many courses. Pay date and amount paid need to store for different courses by different participants. A participant can sit for many exams (excode, title, date) and for each exam, an exam fee (efee) has to pay. Participants can sponsor by their organizations. A sponsor (spcode, spname) organization can have many participants sponsored.

a) Design an ER diagram to capture the above requirements.

(State any assumptions you have that affect your design. Make sure Cardinality Ratios and primary keys are mentioned clearly.) (10 marks)

b) Map the above EER Diagram into a relational schema (10 marks)

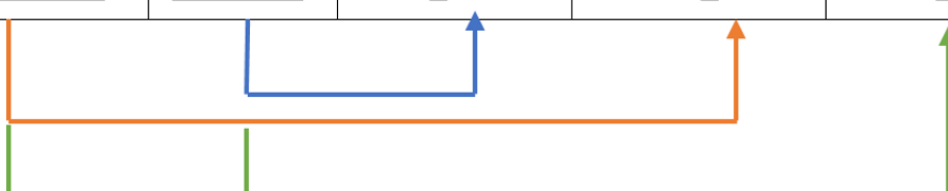
Question Three (20 Marks)

i) What is Normalization? (04 Marks)

ii) Describe 1NF, 2NF and 3NF (No need any examples). (06 Marks)

iii) Consider the below table. Invoice_No and Item_No are Primary Keys

<u>Invoice_No</u>	<u>Item_No</u>	Item_Name	Invoice_Date	Order_Qty



- a. Normalize the data shown in this table to the Third Normal Form (3NF). (You must show step by step) (10 Marks)

Question Four (20 Marks)

Consider the following Tables. One Software Project has many Business Analysts.

SOFTWARE_PROJECT

<u>Project_Code</u>	Project_Name	Duration
P1	Brandix HR Software	6M
P2	MAS HR Software	4M
P3	CEAT Leaves Software	5M

BUSINESS ANALYST

<u>Emp_No</u>	Emp_Name	Emp_Experience	Emp_Salary	Proj_ID
E100	Gayan	02	65000	P1
E200	Waruna	05	98000	P1
E300	Amila	06	125000	P2
E400	Aravinda	01	45000	P3

Based on the above Tables write SQL Queries for the following Requirements. (each question carries 2 marks)

1. Create a database called MIT.
2. Create a table called Software_Project with above fields. Give suitable data types.
3. Create a table called Business Analyst with above fields. Give suitable data types. **Do not assign Proj_ID Column as a Foreign Key.**
4. Alter the Business Analyst table and add Proj_ID as a foreign key that references the Software_Project table.
5. Write SQL statement to insert data to the Business Analyst table. **(Only one insert query is enough)**

6. Display all Business Analysts' details.
7. Display the Employee Number and Employee Name of all Business Analysts whose Salary is greater than 55000.
8. Display all details of Business Analysts whose Salary is between 40000 and 65000 and whose Experience is greater than 03 Years.
9. Display the name of Business Analysts whose names start with 'G'.
10. Can you Drop Software_Project Table? Explain your answer.

Question Five (20 Marks)

Based on the above Tables write SQL Queries for the following Requirements (use Aggregative Functions, Group by, Having, Joining) to do the following: (Each carried 3 marks)

1. Display the Total Salary of Business Analysts in each Software Project.
2. Display the Average Salary of Business Analysts in each Software Project in descending order of Average Salary.
3. Display the Software Projects which have a Total Salary exceeding 96000.
4. Display the Employee Name, Experience and Project Name of all Business Analysts.
5. Display the Employee Name, Experience and Project Name of all Business Analysts who get paid between 45000 and 89000.

Database Security (5 marks)

6. Grant All permission for user1 to Business Analyst table (Suppose user1 already created).
(2.5 marks)
7. Revoke only Select Permission from user1 from Business Analyst table. (2.5 marks)